Arithmetic Pipeline

CSE Sem-4

❖ According to the levels of processing, W. Handler (1977) classify the pipeline into two levels:

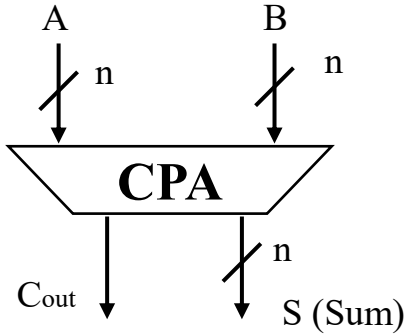
1. **Arithmetic pipelining**
2. **Instruction pipelining**

- **Arithmetic Pipeline:** An arithmetic pipeline divides an arithmetic problem into various sub problems for execution in various pipeline segments. It is used for **fixed point** and **floating-point** arithmetic operations.
- Depending on the functions to be implemented, different pipeline stages in arithmetic unit required different hardware logic.
- As all the arithmetic operations can be implemented with **add** and **shift** operations, the core arithmetic stages require some form of hardware to add or shift.
- Arithmetic or logical shifts can be implemented with highspeed **shift registers**.
- High speed addition requires the use of –
  - **Carry Propagation Adder (CPA)** which adds two numbers and produce an arithmetic sum.
  - **Carry Save Adder (CSA)** which adds three numbers and produce one sum output and a carry output.

- **Carry Propagation Adder (CPA):** Here, carries generated in successive digits are propagated from low end to high end.

For Example, n = 4

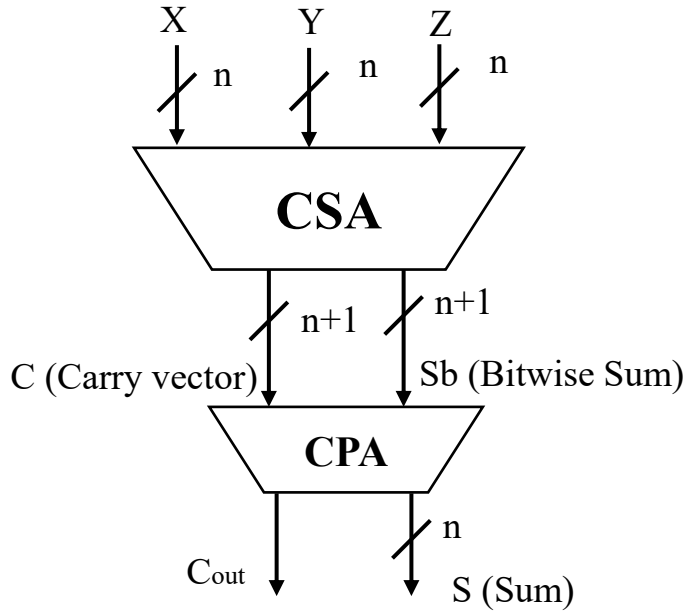
$$\begin{array}{rl} & A = \ 1\ 0\ 1\ 1 \\ + & B = \ 0\ 1\ 1\ 1 \\ \hline & S = 1\ 0\ 0\ 1\ 0 \ = A + B \end{array}$$



- **Carry Save Adder (CSA):** Here, carries are not allowed to propagate but instead are saved in a carry vector. CSA performs bitwise operation simultaneously on all columns of digits to produce (n+1) bitwise sum Sb, and (n+1) carry vector C. MSB of bitwise sum and LSB of carry vector will always be a 0. Sb and C further added using CPA to produce the sum S of 3 input numbers.

For Example, n = 6

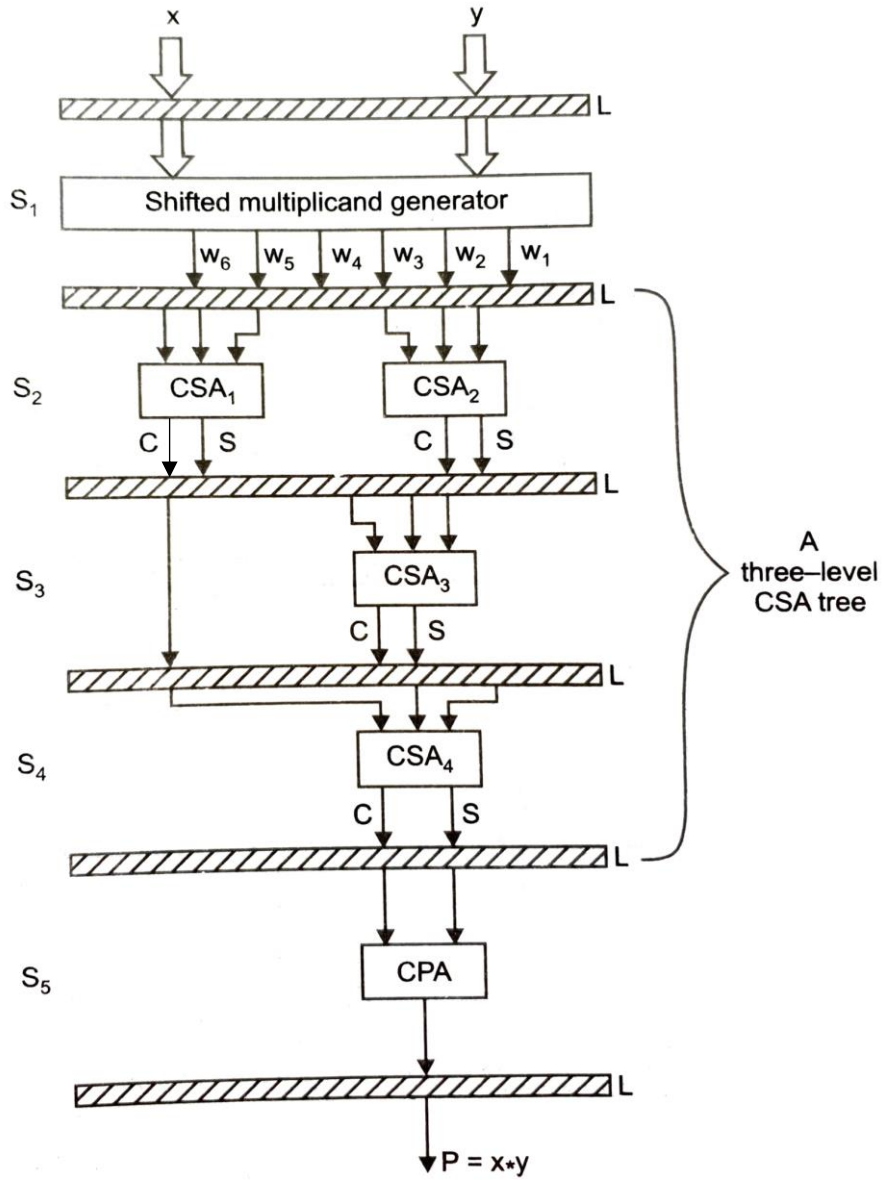
$$\begin{array}{rl} & X = \ 0\ 0\ 1\ 0\ 1\ 1 \\ & Y = \ 0\ 1\ 0\ 1\ 0\ 1 \\ & Z = \ 1\ 1\ 1\ 1\ 0\ 1 \\ \hline & Sb = 0\ 1\ 0\ 0\ 0\ 1\ 1 \\ + & C = 0\ 1\ 1\ 1\ 0\ 1\ 0 \\ \hline & S = 1\ 0\ 1\ 1\ 1\ 0\ 1 \ = Sb + C = X + Y + Z \end{array}$$



❖ Multiplication using pipeline:

Consider a multiplication of two 6-bit numbers X × Y = P, where P is the 12-bit product

- ❑ Stage 1: All 6 partial products (w1 to w6), ranging from 6 bits to 11 bits are generated and passed to stage-2.
- ❑ Stage 2: 6 partial products are added by forming two groups, each having 3 numbers using 2 CSAs, to get 2 sum and 2 carry vectors.
- ❑ Stage 3: The 3 of the above 4 results generated by stage-2 are added by a CSA, and the 4th result is forwarded to stage-4. This CSA also generate 2 results. So finally, 3 results are fed to stage-4.
- ❑ Stage 4: A CSA adds all the 3 results coming from stage-3 and generates 1 sum and 1 carry vector and passes to stage-5.
- ❑ Stage 5: Using a CPA, this stage adds the sum and carry vector received from the previous stage, and generates one number, which is the final product term.




```
      1 1 0 1 0 1 = X
      0 1 0 0 1 1 = Y
-------------------------
      1 1 0 1 0 1 = w1
    1 1 0 1 0 1 0 = w2
  0 0 0 0 0 0 0 0 = w3
0 0 0 0 0 0 0 0 0 = w4
1 1 0 1 0 1 0 0 0 = w5
0 0 0 0 0 0 0 0 0 0 = w6
-------------------------
0 0 1 1 1 1 1 0 1 1 1 1 = P
```

- **Floating point addition or subtraction using arithmetic pipeline:**

X = 0.3214*10^3 and Y = 0.4500*10^2

Floating point addition or subtraction can be performed with four stage pipeline.

Stage 1: Compare the exponents (Two exponents are subtracted to give 3-2 = 1)

Stage 2: Choose the larger exponent as the result exponent and align the mantissas (3 becomes the exponent of result, and the smaller exponent is shifted 1 times to give Y = 0.0450*10^3)

Stage 3: Add or subtract the mantissas (mantissa of X + mantissa of Y = 0.3214 + 0.0450 = 0.3664)

Stage 4: Adjust exponent to normalize the mantissa of result (Normalization means there should be 0 before decimal point, and non-zero value after decimal point. X+Y = 0.3664*10^3)

X = 0.3214*10^3 and Y = 0.4500*10^2

Here, exponents a=3 and b=2
mantissas A = 0.3214 and B = 0.4500

Compare a and b by subtraction:
a – b = 3 – 2 = 1

Exponent of result = 3

X = 0.3214*10^3 and Y = 0.0450*10^3

Mantissa of X + Mantissa of Y
= 0.3214 + 0.0450 = 0.3664

Result = 0.3664*10^3

